

Lab 2 – TrashTag Product Specification

Team Iron

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CS411W

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3. Product Requirements

3.1. Functional Requirements

3.1.1. Litter Reporting (O: Sara)

3.1.1.1. Report Litter Location

- 3.1.1.1.1. The system shall automatically request and gather the user's latitude and longitude coordinates through the browsers built in geolocation tool.
- 3.1.1.1.2. The system shall display the users GPS coordinates on the live-interactive map positioned below the input components of the report screen, which allows for user confirmation before submission.

3.1.1.2. Upload Photos

- 3.1.1.2.1. The system shall incorporate the Camera Capture component, a shared interface component deployed in the detail, report, and event screens to allow the user to either capture a photo directly through the camera or upload from the gallery.
- 3.1.1.2.2. The system shall also incorporate the reusable Camera Capture component described in section 3.1.1.2.1 in the Detail screen as it solicits optional photo upload to verify cleanup status.
- 3.1.1.2.3. The system shall accept form report submission without image upload since photo is optional.

3.1.1.3. Mark Trash Size (lbs)

- 3.1.1.3.1. The system shall display the size input field where the user can enter the estimated weight of the litter in pounds with the purpose of providing more detailed information on the littered site.
- 3.1.1.3.2. The system shall accept submission without the size input field, which is optional.

3.1.1.4. Request Cleanup Verification status

- 3.1.1.4.1. The system shall display an "I cleaned this" button on the detail screen to allow users to mark the reported area as cleaned which will correspondingly update the time and date for the report.
- 3.1.1.4.2. The system shall allow authenticated users to select from options "Still there," "cleaned up," or the "I cleaned this!" buttons on the detail screen to allow users to update the status of the reported area, with

optional photo upload before status status submission.

- 3.1.1.4.3. The system shall display the current status of the litter report on the Trash Detail Screen, reflecting whether the report has been submitted as “I cleaned this”, “Still there” or “cleaned up” through the verification process.

3.1.2. Mapping (O: Sara)

3.1.2.1. View Litter Map

- 3.1.2.1.1. The system shall integrate a mapping library with map tiles for the Litter map screen, with dependencies defined in section 3.3.3.
- 3.1.2.1.2. The system shall display a leaflet popup marker that shows a photo thumbnail of litter and date of report that links to the detail screen defined in Section 3.1.1.4 when clicked.
- 3.1.2.1.3. The system shall allow zoom in and zoom out capabilities to view litter reports in proximity.
- 3.1.2.1.4. The system shall display litter reports as trash icons where yellow is for unverified reports, while red is for verified reports where users stated the trash is still there, and green is for reports that have been cleared.

3.1.2.2. Filter by Status

- 3.1.2.2.1. The system shall allow filtering by report status which would either display or hide reports depending on whether they have been verified as cleaned or still there.

3.1.2.2.1.1. (M: Adam)

The system shall allow filtering by report status which either displays or hides reports depending on their distance and how old they are.

3.1.2.3. Filter by Location

- 3.1.2.3.1. The system shall allow users to filter litter reports by specifying a desired radius in miles, displaying reports within a close proximity range from their current latitude and longitude coordinates.

3.1.3. Client Authorization (O: Adam)

3.1.3.1. Create New Account

- 3.1.3.1.1. The system shall provide the capability for a user to create a new account by submitting a valid email address, username and password.
- 3.1.3.1.2. The system shall verify that the provided email address is not already associated with an existing account.
- 3.1.3.1.3. The system shall verify that the provided username is not already associated with an existing account.
- 3.1.3.1.4. The system shall store user credentials securely upon successful account creation.

3.1.3.2. Login

- 3.1.3.2.1. The system shall provide the capability for a registered user to login using a valid username and password.
- 3.1.3.2.2. The system shall verify the provided credentials by comparing the submitted password (after hashing) with the stored password hash.
- 3.1.3.2.3. The system shall deny access and notify the user if the provided credentials are invalid.
- 3.1.3.2.4. The system shall establish and maintain a user session upon successful login authentication.
- 3.1.3.2.5. The system shall keep the user logged in for their current session once they successfully login.
- 3.1.3.2.6. The system shall verify the provided credentials from the user against all users' data in the database before denying or granting access.

3.1.3.3. View and Edit Profile

- 3.1.3.3.1. The system shall provide the capability for a user to view their stored profile information
- 3.1.3.3.2. The system shall provide the capability for a user to update their profile information, including display name, hometown, and profile image.
- 3.1.3.3.3. The system shall update stored profile information upon the user's successful submission of a profile update.

3.1.3.4. User Types (O: Scott)

- 3.1.3.4.1. The system shall support three different user types.
 - 3.1.3.4.1.1. Standard
 - 3.1.3.4.1.2. Organization
 - 3.1.3.4.1.3. Moderator

3.1.4. Cleanup Tracking (O: Emily)

3.1.4.1. Update Cleanup Status

- 3.1.4.1.1. The system shall provide the user with a status update regarding trash cleanup via a browser-based push notification.
- 3.1.4.1.2. The system shall show the full details of a trash report with a corresponding image gallery.
- 3.1.4.1.3. The system shall fetch the trash report details from the created Django REST API endpoints.

3.1.4.2. Upload Cleanup Proof

- 3.1.4.2.1. The system shall allow the user to upload multiple images of trash to the trash report.

3.1.4.3. Monitor Cleanup Status

- 3.1.4.3.1. The system shall allow the user to manually refresh the trash report page to update the status and view new data.

3.1.5. User Notifications (O: Scott)

- 3.1.5.1. The system shall support a method to show information to the user when the application is not in use. Users can choose which type of notification they like or none at all.
 - 3.1.5.1.1. Browser notifications
 - 3.1.5.1.2. Email

3.1.6. Verification (O: Emily)**3.1.6.1. Verify Accurate Cleanup Status**

- 3.1.6.1.1. The system shall provide the user with the ability to verify trash reports by choosing a trash status ('still there' or 'cleaned up') and uploading optional photos.

3.1.6.2. Verify Cleanup Completion

- 3.1.6.2.1. The system shall provide the user with a browser-based push notification if the verified report has been marked as cleaned and the corresponding trash icon has been removed from the map.
- 3.1.6.2.2. The system shall allow the user to confirm whether the report has been verified.

3.1.6.3. Flag Suspicious Content

- 3.1.6.3.1. The system shall provide the capability for the user to flag suspicious content such as phishing or scam attempts.
- 3.1.6.3.2. The system shall allow the user to flag spam trash reports such as duplicate or false positive reports.

3.1.6.3.3. The system shall allow the user to submit suspicious content to moderators for review.

3.1.6.3.4. The system shall allow the user to submit spam trash reports to moderators for review.

3.1.7. Community Impact (O: Luca)

3.1.7.1. View Impact Statistics

3.1.7.1.1. The system shall provide the user with basic statistics related to the litter reported such as weight in pounds

3.1.7.2. Gamifications/Rewards

3.1.7.2.1. The system shall reward users with points based on their interactions with the application.

3.1.7.2.2. The system shall award users points based on this system.

3.1.7.2.3. For submitting a verified photo of litter with its location, users will earn 100 points.

3.1.7.2.4. For verifying litter is still there, users will earn 50 points.

3.1.7.2.5. For verifying trash has been cleaned up, users will earn 50 points.

3.1.7.2.6. For creating a verified cleanup event that comes to fruition, users will earn 200 points.

3.1.7.2.7. For joining a cleanup event and arriving onsite, users will earn 100 points.

3.1.7.2.8. For completing a verified cleanup event that the user created, users will earn 500 points.

3.1.7.2.9. For completing a verified cleanup event that the user joined, users will earn 200 points.

3.1.7.3. Top Contributors

3.1.7.3.1. The system shall provide users with the option to compete on four leaderboards.

3.1.7.3.1.1. Local weekly.

3.1.7.3.1.2. Local overall.

3.1.7.3.1.3. Global weekly.

3.1.7.3.1.4. Global overall.

3.1.7.3.2. The system shall omit the local leaderboards if users do provide their hometown location.

3.1.7.3.3. The system shall calculate ranking based on points.

3.1.7.3.4. The system shall refresh weekly leaderboards on Sunday at 0:00 EST.

3.1.8. Accessible Resources (O: Adam)**3.1.8.1. Local laws and resources**

- 3.1.8.1.1. The system shall display the following location-based resource types: local environmental laws, recycling guidelines, and contact information for cleanup agencies.
- 3.1.8.1.2. The system shall retrieve resource information based on the user's detected geographic location.
- 3.1.8.1.3. The system shall retrieve information from a curated list of approved resources from government agencies and non-profit organizations .
- 3.1.8.1.4. The system shall display the resource information in structured sections containing heading and body text within the application interface.
- 3.1.8.1.5. The system shall allow users to access resource information without requiring user authentication.

3.1.9. Map Embedding (O: Teddy)**3.1.9.1. Embed Map Widget (Partial)**

- 3.1.9.1.1. The system shall provide an embeddable map view that allows an external organization or government website to display publicly designated trash report locations using the mapping component defined in Section 3.1.2, with only embedding-specific interface constraints differing from the primary map view.
- 3.1.9.1.2. The system shall display each trash report in the embedded map view using the geographic coordinates stored for that report.
- 3.1.9.1.3. The system shall display the selected report's location and cleanup status when a user selects a report marker in the embedded map view.
- 3.1.9.1.4. The system shall restrict the embedded map view to trash report data explicitly designated for public display by an authorized moderator, organization administrator, or government administrator in accordance with the authorization controls defined in Section 3.1.3.

3.1.10. Analytics (O: Teddy)**3.1.10.1. Highlight Predicted Litter Trends (Deferred)**

- 3.1.10.1.1. The system shall calculate litter trend results using stored trash report timestamps and geographic locations.
- 3.1.10.1.2. The system shall group trend results by a system-defined or authorized user-selected time period and a system-defined or authorized user-selected geographic area.
- 3.1.10.1.3. The system shall identify geographic areas in which report counts increase over successive time periods.
- 3.1.10.1.4. The system shall display predicted trend results separately from current reported litter locations.

3.1.10.2. Download Reports (Partial)

- 3.1.10.2.1. The system shall allow an authorized organization or government user to export stored trash report data.
- 3.1.10.2.2. The system shall generate exported report files containing exactly the following fields for each exported report: report identifier, geographic coordinates, submission timestamp, cleanup status, report category, and public display designation status.
- 3.1.10.2.3. The system shall generate the exported report file from the current data stored in the system database at the time of the request.
- 3.1.10.2.4. The system shall deny report export requests from unauthenticated users.

3.1.10.3. View Density - Heavy Litter Analytics (Deferred)

- 3.1.10.3.1. The system shall calculate litter density as the number of active stored trash reports within a defined geographic area, where the geographic area is a system-defined or authorized user-selected radius or boundary around a specified map location.
- 3.1.10.3.2. The system shall generate a density view that distinguishes areas of higher litter concentration from areas of lower litter concentration.
- 3.1.10.3.3. The system shall update density results when new trash reports are added or existing reports are removed from active status.
- 3.1.10.3.4. The system shall allow an authorized user to view density results for a selected map region.

3.1.10.4. Track Event Impact (Deferred)

- 3.1.10.4.1. The system shall allow authorized users to associate cleanup events with one or more trash reports or geographic cleanup areas.
- 3.1.10.4.2. The system shall track changes in report status associated with a cleanup event.
- 3.1.10.4.3. The system shall provide summary metrics for a cleanup event, including the number of reports affected and the number of reports marked cleaned.
- 3.1.10.4.4. The system shall allow authorized users to view event impact information after cleanup event completion.

3.1.10.5. View Cleanup Statistics (Partial)

- 3.1.10.5.1. The system shall provide users with cleanup statistics derived from stored report and cleanup status data.
- 3.1.10.5.2. The system shall display, at minimum, the number of reports submitted, the number of reports marked cleaned, and the number of reports remaining active.
- 3.1.10.5.3. The system shall generate cleanup statistics from data stored in the system database.
- 3.1.10.5.4. In the prototype, cleanup statistics shall be limited to aggregate counts of submitted reports, cleaned reports, and active reports.

3.1.11. Event Coordination (O: Scott)**3.1.11.1. Coordinate Cleanup Events**

- 3.1.11.1.1. The system shall allow users to create events with a given title, description, location, date, and time. The system shall also accept optional photos for event creation.

3.1.11.2. Claim Cleanup Area

- 3.1.11.2.1. The system shall support Organization type user accounts (see 3.1.3.4) marking arbitrary areas (up to 5 square miles) as areas of interest and commitment to keeping clean. They then receive a notification (see 3.1.5) when new trash reports are created in that area.

3.1.11.3. Delegate Cleanup Tasks

- 3.1.11.3.1. The system shall support Organization type user accounts (see 3.1.3.4) assigning Standard user accounts trash reports. The user profile page(see 3.1.3.3) shall show the reports they have been assigned. The system shall allow standard users to remove reports from their profile.

3.1.11.4. Event Participation

- 3.1.11.4.1. The system shall support users viewing on their profile page (see 3.1.3.3) which events they have marked interest or commitment in.
- 3.1.11.4.2. The system shall support Events showing which users have marked interest or commitment in.

3.1.11.5. View Cleanup Events

- 3.1.11.5.1. The system shall support an events list page which shows all events within a threshold distance of the user's current location using GPS (see 3.1.1.1). Threshold is settable by the user in 1, 3, 5, and 10 miles.

3.1.11.6. RSVP to Events

- 3.1.11.6.1. The system shall support users marking interest or commitment to events (see 3.1.11.4.2).

3.1.11.7. Check-In to Events

- 3.1.11.7.1. Users can mark that they are currently attending an event after the event time has started and they are within 500ft of the event location using the device location. (see 3.1.1.1)

3.2. Non-Functional Requirements

3.2.1. Performance Requirements (O: Adam)

3.2.1.1. Map Loading Performance

- 3.2.1.1.1. The system shall display the interactive map within 3 seconds of the user's request.
- 3.2.1.1.2. The system shall render all markers indicating reported trash within 3 seconds of the map loading.

3.2.1.2. Report Submission Performance

- 3.2.1.2.1. The system shall process and store a submitted trash report within 3 seconds of submission.
- 3.2.1.2.2. The system shall upload the image of the trash to the report within 3 seconds.

3.2.1.3. Data Retrieval Performance

- 3.2.1.3.1. The system shall load any trash report's details within 3 seconds of the user selecting it.

3.2.1.4. Concurrent Usage Performance

- 3.2.1.4.1. The systems shall support at least 50 concurrent users performing core actions on the application without any system failures.
- 3.2.1.4.2. The system shall maintain consistent response times while under concurrent usage by less than 100 users.

3.2.2. Usability Requirements (O: Scott)

- 3.2.2.1. The system shall focus on a UI that is best suited for mobile devices, but still maintain comfortable usability on desktop browsers. Examples, but not limited to the following:
 - 3.2.2.1.1. Buttons should be large enough for touch input to be comfortable.
 - 3.2.2.1.2. No functionality should require a user to have a mouse (i.e. no mouse hover or right clicks)

3.2.3. Reliability Requirements (O: Sara)

- 3.2.3.1 The system shall display an error alert if there is failure to retrieve data on any screen.
- 3.2.3.2 The system shall redirect any unauthenticated users to the account screen and restrict access to protected screens until the user is successfully logged in.

3.2.4. Security Requirements (O: Teddy)

3.2.5. Availability Requirements (O: Emily)

- 3.2.5.1. The system shall be operational and available online 99.9% of the time for all registered and non-registered users, except for required maintenance periods.

3.2.6. Maintainability Requirements (O: Emily)

- 3.2.6.1. The system shall have well-documented code according to industry coding standards.
- 3.2.6.2. The system shall implement version control via GitHub and make use of branching practices to update existing features and implement new features.

3.2.7. Scalability Requirements (O: Scott)

- 3.2.7.1. The system shall support up to X simultaneous users.

3.3. Constraints

3.3.1. Technology Constraints (O: Adam)

3.3.1.1. Web Application Architecture

3.3.1.1.1. The system shall be implemented as a web-based application that is accessible through modern web browsers.

3.3.1.1.2. The system shall follow a client-server architecture for user requests and data processing.

3.3.1.2. Backend and Database Technologies

3.3.1.2.1. The system shall utilize a server-side framework to manage application logic.

3.3.1.2.2. The system shall use a relational database to store user data, trash reports, and associated metadata.

3.3.1.3. Geolocation and Mapping Integration

3.3.1.3.1. The system shall integrate a mapping service to display trash report locations on a map.

3.3.1.3.2. The system shall rely on browser-based geolocation abilities to identify the location of the user.

3.3.1.3.3. The system shall use the retrieved user location data to display trash reports near the location of the user.

3.3.1.4. Media Handling

3.3.1.4.1. The system shall support image uploads and image storage for trash reports.

3.3.1.4.2. The system shall support capturing and uploading a new image while the application is open.

3.3.2. Platform Constraints (O: Scott)

3.3.2.1. The application shall be runnable in any modern browser.

3.3.2.2. The system requires installation on a single computer capable of running python with Django and npm.

3.3.2.2.1. Python version: 3.12.3

3.3.2.2.2. Django version: 6.0.1

3.3.2.2.3. Npm version: 9.2.0

3.3.2.3. The system requires acceptance of inbound HTTPS traffic at a static address. Domain name recommended.

3.3.2.4. The system requires a web service capable of serving static content and dynamic content through Python.

3.3.3. Implementation Constraints (O: Sara)

3.3.3.1. The system shall use the React Leaflet library for map functionality and OpenStreetMap tiles for the visual map data to allow for accurate location rendering.

3.3.3.2. The system shall use the browsers built in navigator.geolocation tool to obtain users GPS coordinates.

3.3.4. External Dependencies (O: Scott)

- 3.3.4.1. The system shall make use of the OpenStreetMap tile API to display mapping data.

3.3.5. Legal and Privacy Constraints (O: Luca)

- 3.3.5.1. The system shall require users to agree to an End User Licensing Agreement (EULA).